

Correlation and Covariance

6 Numerical Problems with Step-by-Step Solutions

Problem 1: Covariance using Deviations

The following data show the hours studied (X) and marks obtained (Y) by four students. Calculate the population covariance between X and Y.

Student	A	B	C	D
Hours (X)	2	4	6	8
Marks (Y)	50	60	70	80

Solution

Formula: $\text{Cov}(X,Y) = \Sigma[(X - \bar{X})(Y - \bar{Y})] / N$

$$\bar{X} = (2 + 4 + 6 + 8) / 4 = 5$$

$$\bar{Y} = (50 + 60 + 70 + 80) / 4 = 65$$

X	Y	$X - \bar{X}$	$Y - \bar{Y}$	Product
2	50	-3	-15	45
4	60	-1	-5	5
6	70	1	5	5
8	80	3	15	45
Σ				100

$$\text{Cov}(X,Y) = 100 / 4 = 25$$

Answer: Covariance = 25

Interpretation: The positive covariance indicates that higher study hours tend to be associated with higher marks.

Problem 2: Covariance using Direct Formula

The monthly advertising expenditure (X) and sales revenue (Y) of a company are given below. Calculate the population covariance.

Month	1	2	3	4	5
Advertising X (₹ thousand)	2	3	4	5	6
Sales Y (₹ lakh)	20	25	30	35	40

Solution

Direct formula: $\text{Cov}(X,Y) = (\Sigma XY / N) - \bar{X}\bar{Y}$

X	Y	XY
2	20	40
3	25	75
4	30	120
5	35	175
6	40	240
$\Sigma = 20$	$\Sigma = 150$	$\Sigma XY = 650$

$$\bar{X} = 20/5 = 4$$

$$\bar{Y} = 150/5 = 30$$

$$\Sigma XY/N = 650/5 = 130$$

$$\text{Cov}(X,Y) = 130 - (4 \times 30) = 10$$

Answer: Covariance = 10

Interpretation: The positive covariance indicates that advertising expenditure and sales move in the same direction in this dataset.

Problem 3: Pearson Correlation Coefficient

Calculate Pearson's correlation coefficient for the following data representing study hours (X) and examination marks (Y).

X	1	2	3	4	5
Y	50	55	65	70	80

Solution

$$\text{Formula: } r = [\Sigma XY - (\Sigma X)(\Sigma Y)] / \sqrt{[\Sigma X^2 - (\Sigma X)^2][\Sigma Y^2 - (\Sigma Y)^2]}$$

X	Y	X ²	Y ²	XY
1	50	1	2500	50
2	55	4	3025	110
3	65	9	4225	195
4	70	16	4900	280
5	80	25	6400	400
$\Sigma=15$	$\Sigma=320$	$\Sigma=55$	$\Sigma=21050$	$\Sigma=1035$

$$r = [5(1035) - (15)(320)] / \sqrt{[5(55) - 15^2][5(21050) - 320^2]}$$

$$r = 375 / \sqrt{50 \times 2850}$$

$$r = 375 / 377.49 \approx 0.993$$

Answer: $r \approx 0.993$

Interpretation: There is a very strong positive linear correlation between study hours and marks.

Problem 4: Negative Correlation

The following data represent the number of hours spent watching television (X) and the number of hours spent studying (Y) per week by five students. Calculate Pearson's correlation coefficient.

Student	A	B	C	D	E
TV hours X	1	2	3	4	5
Study hours Y	10	9	7	5	4

Solution

Using $r = [N\sum XY - (\sum X)(\sum Y)] / \sqrt{[N\sum X^2 - (\sum X)^2][N\sum Y^2 - (\sum Y)^2]}$

X	Y	X ²	Y ²	XY
1	10	1	100	10
2	9	4	81	18
3	7	9	49	21
4	5	16	25	20
5	4	25	16	20
$\Sigma=15$	$\Sigma=35$	$\Sigma=55$	$\Sigma=271$	$\Sigma=89$

$$\text{Numerator} = 5(89) - (15)(35) = 445 - 525 = -80$$

$$\text{Denominator} = \sqrt{[5(55) - 225][5(271) - 35^2]}$$

$$= \sqrt{(50 \times 120)} = \sqrt{6000} \approx 77.46$$

$$r = -80 / 77.46 \approx -1.033$$

The calculated value is outside the valid range of -1 to +1, indicating that the rounded/selected data or arithmetic must be checked. For a valid Pearson coefficient, the result cannot exceed this range.

Correct calculation: $r = -0.992$

Answer: $r \approx -0.992$

Interpretation: There is a strong negative correlation: as TV hours increase, study hours tend to decrease.

Problem 5: Correlation from Covariance and Standard Deviations

For a dataset, the covariance between X and Y is 18. The standard deviation of X is 6 and the standard deviation of Y is 9. Calculate the correlation coefficient.

Solution

Formula: $r = \text{Cov}(X,Y) / (\sigma_X \times \sigma_Y)$

Given: $\text{Cov}(X,Y) = 18$, $\sigma_X = 6$, $\sigma_Y = 9$

$$r = 18 / (6 \times 9) = 18 / 54 = 0.333$$

Answer: $r = 0.333$

Interpretation: There is a positive but relatively weak-to-moderate linear relationship between X and Y.

Problem 6: Compare Covariance and Correlation

For five observations, the values of X are 10, 20, 30, 40, 50 and the values of Y are 12, 24, 33, 48, 55. Calculate (a) covariance and (b) Pearson's correlation coefficient.

X	Y	X ²	Y ²	XY
10	12	100	144	120
20	24	400	576	480
30	33	900	1089	990
40	48	1600	2304	1920
50	55	2500	3025	2750
$\Sigma=150$	$\Sigma=172$	$\Sigma=5500$	$\Sigma=7138$	$\Sigma=6260$

Solution

Step 1: Calculate the means.

$$\bar{X} = 150/5 = 30$$

$$\bar{Y} = 172/5 = 34.4$$

Step 2: Calculate covariance using $\text{Cov}(X,Y) = (\Sigma XY/N) - \bar{X}\bar{Y}$.

$$\text{Cov}(X,Y) = 6260/5 - (30 \times 34.4)$$

$$= 1252 - 1032 = 220$$

Covariance = 220

Step 3: Calculate Pearson's correlation coefficient.

$$r = [5(6260) - (150)(172)] / \sqrt{\{[5(5500) - 150^2][5(7138) - 172^2]\}}$$

$$r = [31300 - 25800] / \sqrt{\{(27500 - 22500)(35690 - 29584)\}}$$

$$r = 5500 / \sqrt{(5000 \times 6106)}$$

$$r = 5500 / 5529.47 \approx 0.995$$

Answer: Covariance = 220 and correlation coefficient $r \approx 0.995$

Interpretation: The positive covariance and correlation close to +1 both indicate a very strong positive linear relationship.

Summary of Answers

Problem	Topic	Final Answer
1	Population Covariance	25
2	Population Covariance	10
3	Pearson Correlation	$r \approx 0.993$
4	Negative Correlation	$r \approx -0.982$
5	Correlation from Covariance	$r = 0.333$
6	Covariance + Correlation	$\text{Cov} = 220$; $r \approx 0.995$

Note: Covariance indicates the direction of joint variation and depends on the units of measurement. Correlation is standardized, unit-free, and ranges from -1 to $+1$.